

Spatio-temporal Gait Characterization from a Wrist-Worn Inertial Measurement Units in Daily Life

Prerna Advani (pa2671), Jordan Jomsky (jj3379), Kate Lassiter (kal2257), Jake Thor Meyer (jtm2217), Karl Rodrigo Garcia-Meza (krg2150)

December 19, 2024

Executive Summary

Background:

Gait analysis is a crucial component in the assessment and monitoring of various health conditions, including neurological disorders, musculoskeletal issues, and recovery from injuries. Traditional methods often require specialized equipment and controlled environments, limiting their applicability in real-world settings. This project aimed to leverage the ubiquity and convenience of wrist-worn inertial measurement units (IMUs) to perform spatio-temporal gait characterization in daily life activities. The data from a sensor attached to the lower back served as the ground truth for labeling the data, while the wrist-worn sensor data was used for algorithm development and model training. This approach can significantly enhance the accessibility and practicality of gait analysis, providing continuous monitoring and early detection of gait abnormalities in a variety of real-world contexts.

Objective:

The primary goal of this project was to develop and validate algorithms that can accurately characterize spatio-temporal gait parameters using data from a wrist-worn IMU. Specific tasks included:

- Applying open-source lower back gait detection algorithms to partially labeled lower back data to validate the best approach for generating labels for the whole dataset.
- Using generated labels from the lower-back model(s) to evaluate the feasibility of classifying subject activities from wrist-based data via heuristic approaches as well as a custom deep convolutional neural network (DCNN).
- Leveraging the results from the steps above to feed models that accurately characterize subjects' gait, focusing on walking speed.

Findings:

The teams' efforts yielded promising results, as we successfully implemented models to detect gait information from lower-back data, which informed the development of wrist-based techniques better suited for real-world applications. Wrist-based models showed strong performance in identifying walking instances and estimating walking speed, which can offer valuable insights for assessing medical conditions and recovery in a healthcare setting. While further refinement is needed, this work underscores the feasibility of using wrist-worn IMUs for gait analysis and lays a foundation for future research into scalable, practical solutions.

1 Introduction

Analyzing gait patterns has become an essential tool in healthcare, offering insights into various medical conditions and recovery processes. For example, using gait speed estimation alone, medical providers are able to characterize orthopedic diseases, predict the risk of falling, and quantify impacts of intervention (Soltani et al., 2020). However, traditional analysis methods often require specialized equipment and controlled settings, making it more difficult to generalize results to real-world applications. This research aims to develop and compare algorithms for accurately classifying activities (i.e., high activity, low activity) and characterizing spatio-temporal gait parameters, such as step length and walking speed, using data from wrist-worn inertial measurement units (IMUs) in everyday settings.

The rising popularity of wearable devices, particularly those worn on the wrist, has opened new avenues for continuous health monitoring. Recent studies have shown promising results in using wrist-worn accelerometers for gait analysis, and researchers have developed machine learning algorithms capable of detecting steps with reasonable accuracy in non-laboratory environments. For instance, Small et al. (2023) developed a machine learning algorithm for step detection using wrist-worn accelerometers that achieved a mean absolute percent error (MAPE) of 12.5%, a rate comparable to the ~10% MAPE achieved via step counting algorithms during controlled walking and running activities in a different study by Femiano et al. (2022).

Additionally, large-scale studies utilizing extensive datasets have enhanced our ability to recognize human activities, including gait-related movements, through self-supervised learning techniques. Yuan et al. (2024) utilized self-supervised learning on a large dataset of 700,000 person-days of wearable data to enhance human activity recognition, including gait-related activities. The authors noted that model performance improved significantly with increased access to labeled data, particularly in settings with initially limited labeled examples, as more labeled data allowed for better training and accuracy across activity recognition tasks.

Despite these advancements, accurately extracting gait parameters from wrist-worn sensors remains challenging. The complexity of arm movements and the variety of daily activities make it difficult to distinguish walking patterns from other hand motions, especially in individuals with altered gait. Brand et al. (2022) highlighted this issue, noting that the wrist's many degrees of freedom introduce variability that complicates the task of distinguishing actual gait events from other non-gait hand movements (e.g., brushing teeth), especially in populations with gait impairments. At the same time, there have been promising advances in activity classification: Kulge et al. (2023) validated wrist-based gait sequence detection algorithms against reference systems in diverse cohorts, including individuals with Parkinson's disease, hip fracture recovery, and chronic obstructive pulmonary disease. While performance metrics such as sensitivity and specificity for wrist-worn sensors were slightly lower compared to those for lower-back sensors, their ability to detect gait sequences with high specificity highlights their applicability in real-world scenarios.

Expanding the scope of sensor-based gait analysis, Iluz et al. (2014) developed an objective tool for detecting missteps in patients with Parkinson’s disease using lower-back sensors, demonstrating high sensitivity and specificity in real-world conditions and thereby enhancing fall risk assessment. Additionally, Paraschiv-Ionescu et al. (2019) devised a trunk-fixed accelerometer algorithm for locomotion and cadence detection in children with cerebral palsy, showcasing robust performance across a range of atypical gait patterns in daily life-like conditions. These studies collectively underscore the versatility and potential of inertial sensors in diverse populations and settings. Building upon these efforts, our research aims to refine wrist-based gait analysis by leveraging both machine learning and sensor fusion techniques to address challenges such as variability in arm movements and to enhance the accuracy of spatio-temporal gait parameter estimation in everyday settings.

To achieve this, we analyzed raw sensor data collected from lower-back and wrist IMUs worn by 28 participants during various daily activities. The accelerometers in these IMUs captured acceleration along the X, Y, and Z axes, reflecting movement intensity and direction, and serving as inputs to our models. The lower-back sensor, widely regarded as the gold standard for gait detection, was used to derive ground truth data for both activity classification and gait analysis. Meanwhile, the wrist-worn sensor data was used to develop and train algorithms to generalize activity recognition and gait parameter estimation to real-world settings, where wrist-worn devices are more practical and commonly used.

In the following sections, we describe the dataset and methods used to: (1) derive ground truth labels from the lower-back sensor; (2) classify participants’ activities from wrist data using heuristic and deep learning approaches; and (3) estimate gait parameters, with a specific focus on walking speed.

2 Data

2.1 Dataset Description

In total, the dataset monitors data for 28 subjects, providing detailed temporal data across a variety of physical states and activities, which are useful for activity recognition and analysis over time. These subjects were tracked over the course of three days to a month. A key metadata file, `demographics.csv`, includes personal details of each subject, such as their age, gender, height, and the duration of their measurement period. In Figure 1, we observed the demographics of our subject population. Our subjects skewed older and more female. Thus, our subjects tended to be less tall and weigh less.

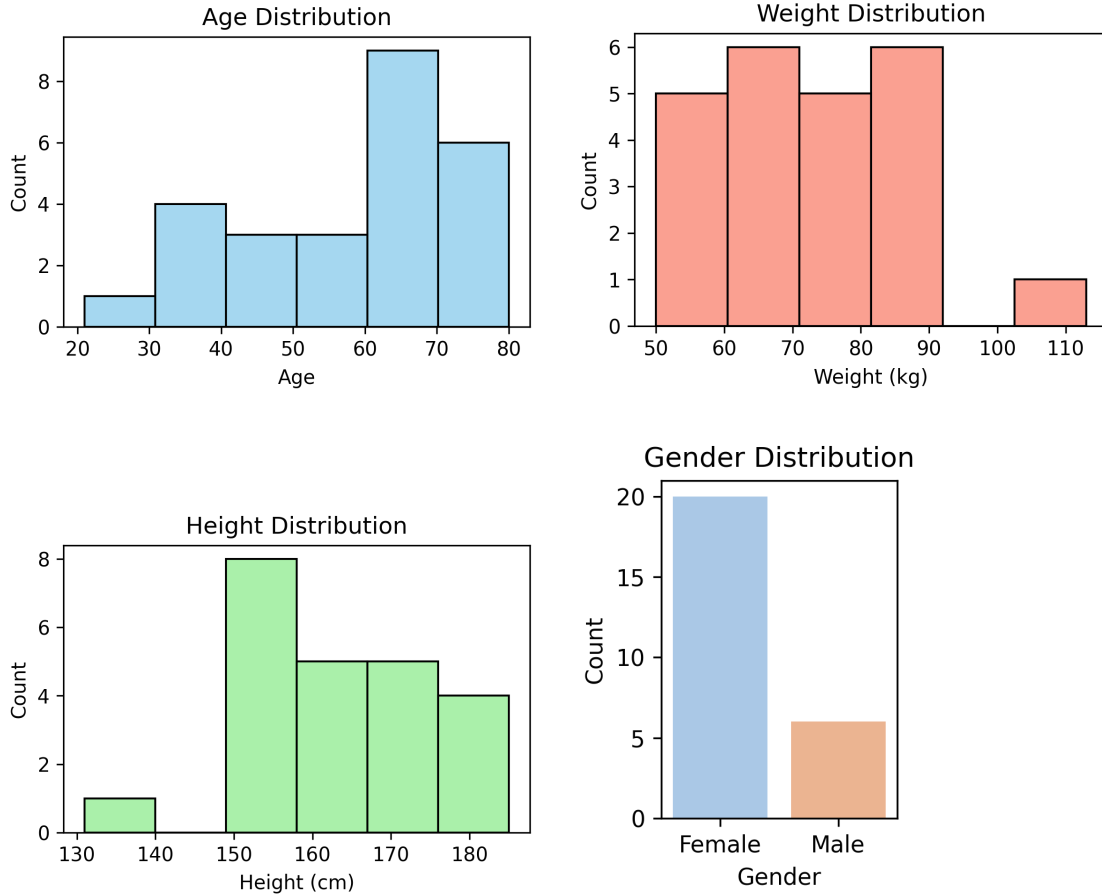


Fig. 1: Subject Demographics

Our data was collected from two primary groups of sensors: lower back and wrist sensors. These sensors capture the acceleration and rotation of the subject. These measurements are taken with a frequency of 50-100 Hz, creating a data-rich signal to process. This dataset is originally captured as Open Matrix (OMX) files and can further be processed into CSV labels. OMX files, based on the HDF5 standard, store matrices and metadata, specifically lower back data in this context. CSV files are central to the dataset and are used for various purposes, such as raw data storage, activity classification, and subject metadata.

One key CSV file, `combined_ax6_df.csv`, is found in every subject folder and contains unlabeled accelerometer and gyroscope data from wrist sensors. It records attributes like accelerometer (X, Y, Z), gyroscope (X, Y, Z), and temperature readings. Another important CSV file, `mapped_df.csv`, includes similar fields and labeled activities (e.g., walking, sitting) for a small subset of subjects. These labels are used to map physical activities over time. We observed a significant class imbalance for every subject as shown in Fig. 1.

Low Activity		High Activity	
Label	% of Total	Label	% of Total
not_worn	30.76%	walking	3.36%
sitting	26%	shuffling	0.80%
lying	33.17%	stair_walking	0.17%
standing	5.74%	cycling	0.01%

Fig 2: Frequency of Activity Class in mapped_df.csv

There are also classification files such as combined_classification_df.csv, which contain activity classifications for specific time intervals, providing insights into gait analysis, steps taken, energy expenditure, and other metrics. For lower back data, files named after each subject, such as [subject name]-[code]-[date]-[date].csv, contain timestamped accelerometer, gyroscope, and magnetometer readings. This file has a mapped variant that contains the ground truth labels for some of the subjects, as we had incomplete data on the activities of all of these subjects. In Figure 2, we show the different activities that were recorded in this ground truth as well as the groupings in the high versus low activity categories.

One challenge we faced was the size of our datasets, with most subjects' files exceeding 10GB, which made data ingestion difficult on resource-limited machines. To address this, we initially considered chunking the data to process smaller portions at a time. Ultimately, we resolved the issue by upgrading our workspaces to instances with larger memory capacities.

2.2 Visualizations

We began exploring the data by calculating key features like velocity, position, and acceleration over time given accelerometer data.

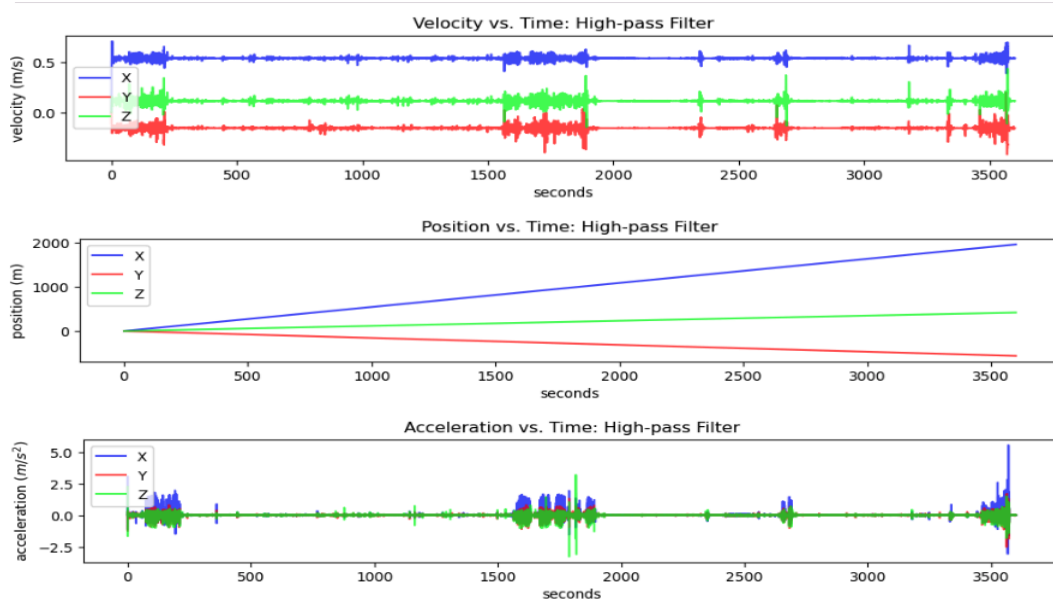


Fig. 3: Butterworth filter, high-pass, position velocity and acceleration

Then, we experimented with signal processing packages. In Figure 3, we displayed a Butterworth Digital and Analog Filter Design using a high-frequency filter. From this figure, we can clearly see that the acceleration signal offers the highest potential for discriminative analysis for activity classification due to the stark difference between high and low activity acceleration compared to position and velocity.

Given that our aim was to detect instances of participants' walking, referred to as 'walking bouts', focusing on acceleration seemed most logical and reduced the number of dimensions for modeling tasks considerably.

3 Methodology

As shown in Fig. 4, our approach for achieving wrist-based activity classification and gait parameter estimation involved the following steps:

- Applying open source lower back gait detection algorithms to partially labeled lower back data to validate the best approach for generating labels for the whole dataset.
 - These algorithms include a baseline machine learning-based model and a more advanced, 'final model' developed using a recent lumbar data analysis library.
- Using generated labels from the better of the two lower-back models to evaluate the feasibility of classifying subject activities (i.e., walking bout detection) from wrist-based data via heuristic approaches as well as a custom deep convolutional neural network (DCNN).
- Leveraging the results from the steps above to feed models that accurately characterize subjects' gait, focusing on walking speed.

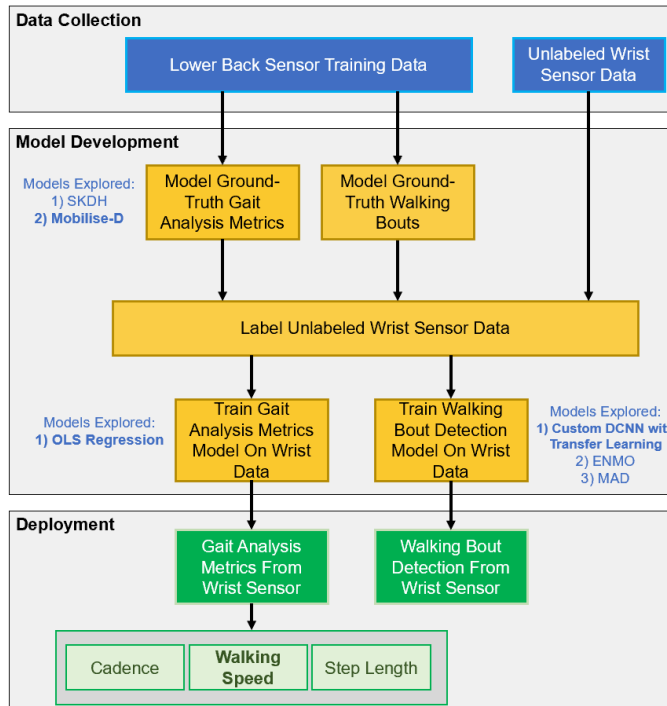


Fig. 4: Project Pipeline

Baseline Lumbar Model - SKDH

3.1 Analytic Methods

For our baseline model, we used an established, open-source API called SciKit Digital Health (SKDH). Introduced in Adamowicz et al. (2022), this package contains a pre-trained model for gait lumbar classification that we can apply to our data out-of-the-box. The gait detection model uses a Light Gradient Boosting Machine (LightGBM) classifier to identify gait bouts from lumbar accelerometer data, processed in non-overlapping 3-second windows. LightGBM is an open-source machine learning framework that improves the speed and efficiency of tree-based algorithms, like eXtreme Gradient Boosting (XGBoost).

First, the model preprocesses input data by verifying and adjusting the sampling frequency to either 50 Hz or 20 Hz, applying an anti-aliasing filter if needed. The acceleration signals are then bandpass-filtered (0.25-5 Hz) to isolate the frequency range most indicative of gait. Features are extracted from these filtered signals on a per-window basis, which includes metrics optimized for gait classification. To enhance prediction reliability, the model attempts to maximize F1 score to the classifier's output to reduce false positives. Results provided in the form of the gait bout start and stop times as indices. This model enables effective segmentation of continuous accelerometer data, facilitating accurate gait action classification assessments.

3.2 Validation and Testing

To evaluate the effectiveness of the model, we used two types of evaluation methods:

Validation Using Ground Truth Data: The model was evaluated by comparing the performance of our algorithms on wrist data versus the ground truth lower back data to assess the viability and accuracy of wrist-worn IMUs for continuous gait monitoring in everyday life. We used time series-aware validation methods, with splitting performed by chunking the data into windows, randomly selecting partitions and training an XGBoost classification model on the extracted features.

Standardized Metrics: The model was benchmarked using subject-level metrics such as F1, Recall, and Precision across 13 subjects.

3.3. Results

In Figure 5, we sought to understand the performance of SKDH's LGBM model. This figure displayed the outcomes of one subject's prediction using the LGBM model versus the ground truth over time.

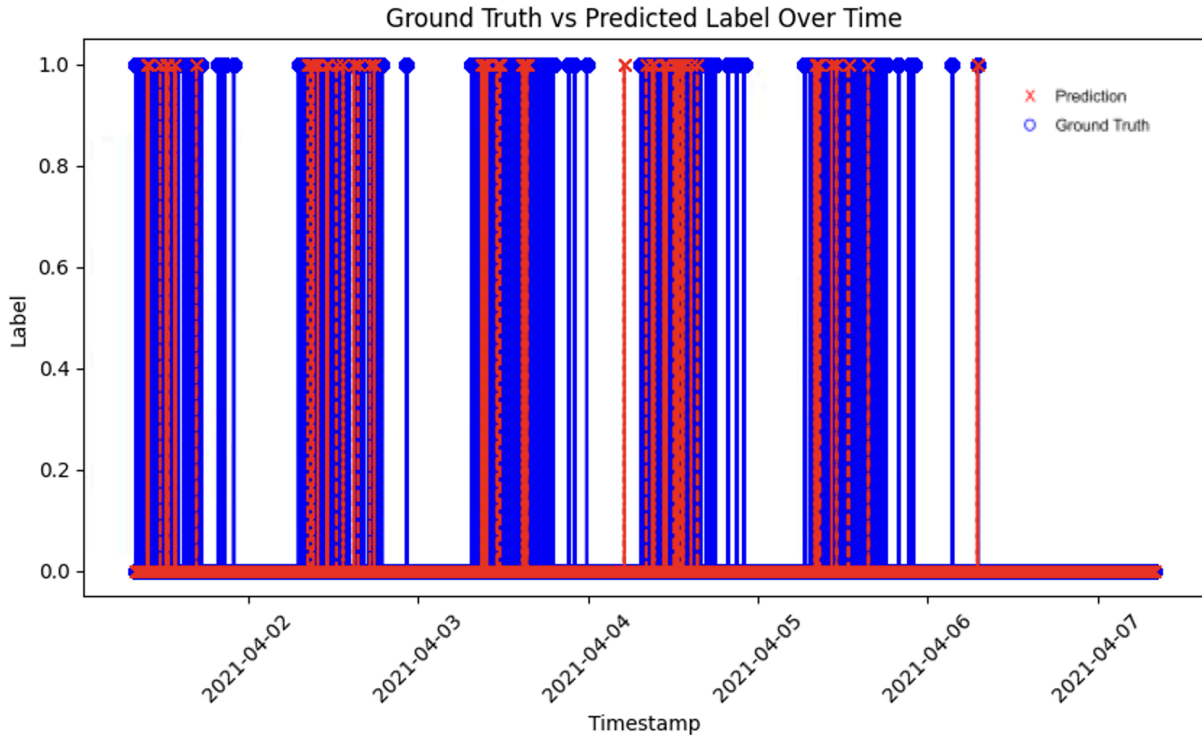


Fig. 5: LGBM results

The results show that while LGBM can detect instances of walking, there is significant room for improvement. In terms of metrics, the F1 and precision were both 0 for the model, we suspect due to significant class imbalance and sub-optimal model performance, further exacerbated by the missingness and complexity in our time-series data.

Final Lumbar Model - Mobilise-D

3.4 Analytic Methods

For our final lumbar model, we relied upon the Mobilise-D library, which came from a 5-year project funded by the Innovative Medicines Initiative (IMI) to help monitor gait for people with various mobility issues (Rochester et al., 2020). This package offered a collection of APIs that can be used to build a pipeline for gait analysis using IMU worn on the lower back. The library offers two different models based on two papers.

Iluz:

The first model that we tried implemented a method described by Iluz et al. (2014), which attempted to identify missteps in a population of patients with Parkinson's Disease. The researchers observed that during normal walking, the acceleration signal exhibited a repeatable pattern within frequencies of 0.5–3.0 Hz. This observation formed the basis for the model, which accounted for disruptions caused by tremors, changes in gait, navigating obstacles, or sensor placement by band-pass filtering the signal to isolate only frequencies in this range. The

algorithm processed the filtered signal using 5-second sliding windows on the vertical and anterior-posterior axes. Within each window, the data was convolved with a single cycle of a 2 Hz sinusoidal signal, modeling the periodic nature of a typical gait cycle. Peaks in the resulting signal (local maxima) therefore correspond to individual gait cycles. Finally, windows with 2–15 detected steps were classified as gait. This range aligns with typical gait frequencies: 0.5 Hz represents a stride every two seconds (2 steps in 5 seconds), and 3.0 Hz represents three steps per second (15 steps in 5 seconds).

Ionescu:

The second model was based on a paper focused on identifying heel-strike events from signals produced by children with cerebral palsy (Paraschiv-Ionescu et al., 2019). The algorithm first preprocessed the signal, taking the norm of the 3-dimensional acceleration vectors. The acceleration norm was then resampled to 40 Hz, detrended, and low-pass filtered using a cut-off of 3.2 Hz. The signal was filtered once in the forward direction and then reversed and filtered again in the reverse direction to prevent any phase shifts from the filtering process. A continuous wavelet transform and Savitsky-Golay filter (which finds the average every three samples) were then applied to the signal to further smooth it. Finally, a threshold of 0.1g was used for a rule-based approach to identify the heel-strike events by checking the amplitude of the processed signal. A dynamic time threshold starting at 3.5 seconds (which changed depending on the average duration of the previous steps) was also used to help identify whether or not the subject was still walking or not. After each period of walking was identified, only the walking bouts of at least four steps were kept.

We tested both of these models to identify gait sequences to use as the ground truth for our wrist model.

3.5 Validation and Testing

Similarly to the baseline SKDH model, we used the validation of predicted activity labels versus ground truth to evaluate our models in addition to obtaining standardized metrics such as F1 and precision. Both models, Iluz and Ionescu, were tested under identical preprocessing conditions, though Iluz was run on data from 13 subjects while Ionescu was run on data from 15 subjects. Due to the Iluz algorithm's strict requirements for sufficient activity within a given time period, two subjects were excluded from its execution.

3.6. Results

Upon validating our predicted labels versus ground truth, we found that the Ionescu model outperformed by detecting most extended walking bouts. In Figure 6, we showed a raw acceleration signal. Clearly, the noisy periods indicate high activity and the less noisy indicate low activity. We expect an algorithm to capture this accurately to effectively work as a ground truth for our wrist model. In Figure 7, on visual inspection, we saw great performance from the Ionescu model on our data. In fact, in comparison to the given ground truth labels shown in

Figure 8, we saw a much smoother representation of high versus low activity in the signal in the Ionescu model's predictions than in the ground truth label itself. While it may seem that the ground truth labels have more granularity, in reality, this is due to missingness in the given data as shown in Figure 9. However, we note that there was a tendency to miss some of the shorter walking periods and some inconsistency with accurately pinpointing the start/end of the walking bout in the Ionescu model.

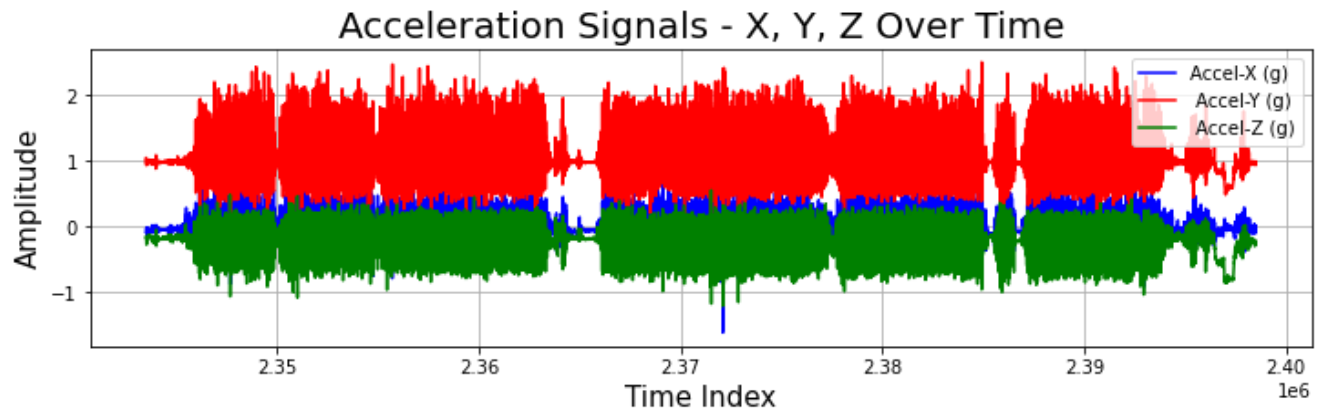


Fig. 6: Acceleration Signals

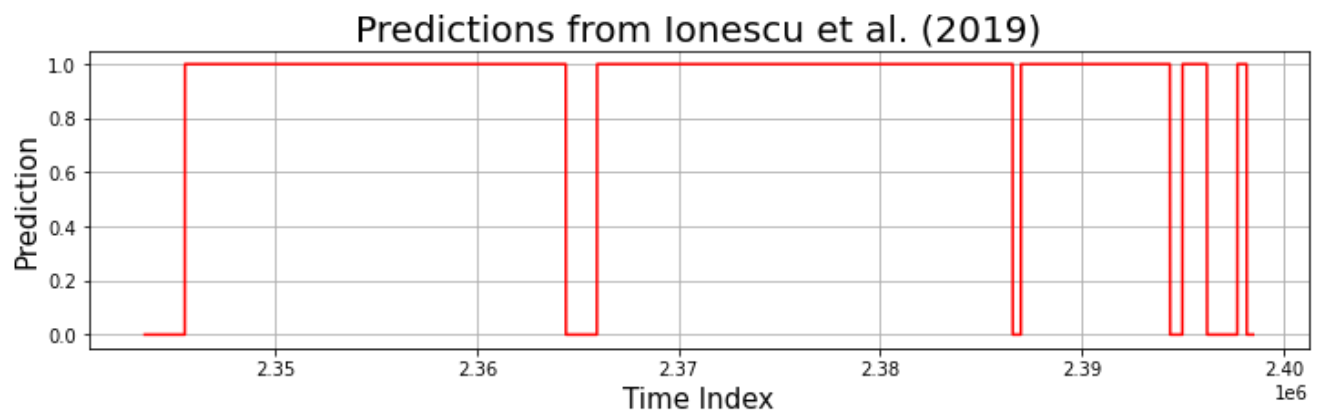


Fig. 7: Predicted Values

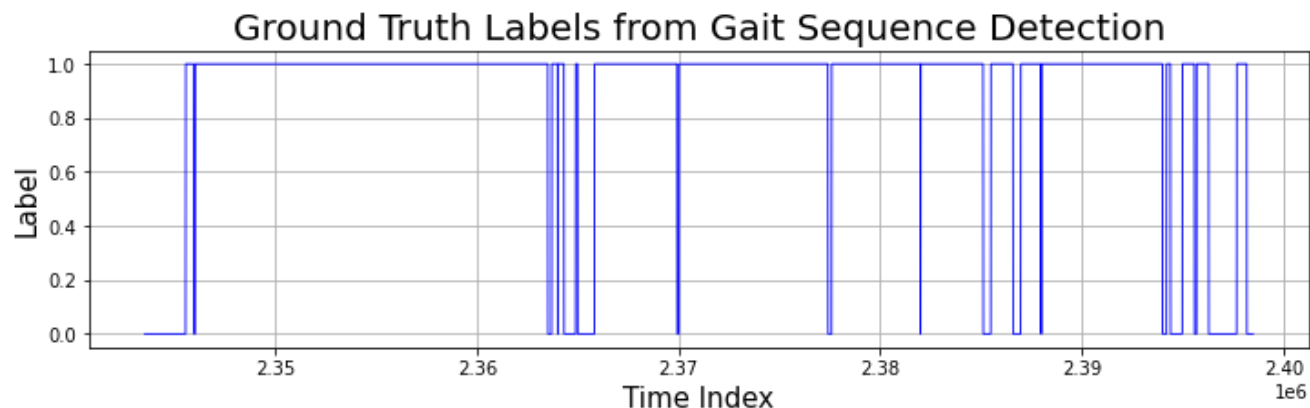


Fig. 8: Ground Truth Labels

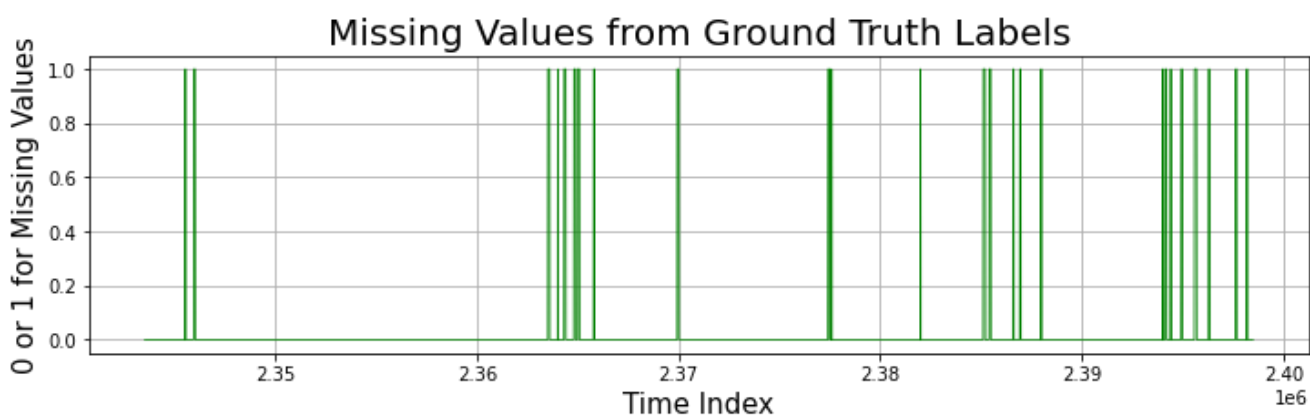


Fig. 9: Missing Values

We found that the Ionescu model demonstrated the best accuracy (F1-score: 0.41) in adult gait prediction. Its rule-based dynamic time threshold (3.5 seconds) was critical for handling variable gait paces. Interestingly, the model seemed to perform well for six subjects out of the fifteen that were available with demographic data, averaging an F1 score of 0.78. However, for the majority of subjects, the model failed to accurately identify the gait sequences from the signal. Overall, the available data yielded F1 scores ranging from 0.07 to 0.83. We were unable to ascertain a specific reason for this lopsided performance despite searching for patterns in the demographic data.

Position	Model	F1	Precision	# of Subjects
Lower Back	SKDH	0	0	13
	Mobilise-D - Iluz	0	0	13
	Mobilise-D - Ionescu	0.41	0.40	15

Fig. 10: Model Metrics for Lower Back

Baseline Wrist Model - MAD & ENMO

3.7 Analytic Methods

Having obtained activity classification and gait analysis labels from the better-performing Ionescu lower-back model, the team shifted focus to wrist-based data for these tasks. For activity classification, we began by evaluating two heuristic methods, Euclidean Norm Minus One (ENMO) and Mean Amplitude Deviation (MAD), as baseline models.

ENMO measures dynamic movement intensity by calculating the L2 norm of the raw accelerometer signal (X, Y, Z) and subtracting 1g to remove the effect of gravity (Bakrania et al., 2016):

$$ENMO = \max\left(\sqrt{x_i^2 + y_i^2 + z_i^2} - 1, 0\right)$$

The higher the ENMO value, the greater the activity intensity is estimated to be. To derive binary activity labels (walking/higher-intensity = 1; sedentary = 0), we computed a rolling average of ENMO over 5-second windows (500 rows) and experimented with thresholds of 0.035g, 0.05g, 0.07g, and 0.1g, where a 5-second average exceeding these thresholds was classified with the label 1.

A similar process was followed for using MAD as our baseline model. MAD quantifies signal variability as the mean absolute deviation from the rolling mean of the L2 norm (Bakrania et al., 2016).

$$MAD = \frac{1}{n} \times \sum_{i=1}^n |r_i - \bar{r}|$$

where $r_i = \sqrt{x_i^2 + y_i^2 + z_i^2}$ and $\bar{r}$ is the mean vector magnitude within a 5 second window. Like ENMO, higher MAD values indicate greater activity intensity. Binary labels were similarly derived by applying thresholds to the MAD signal.

3.8 Validation and Testing

To evaluate the performance of ENMO and MAD, we compared their predicted labels with the ground truth activity labels obtained from the Ionescu lower-back model. We achieved this by developing a script that looped through the lower-back model's gait and activity classification predictions for each subject, mapping walking periods back to the corresponding wrist data. Once the labels were transferred, we evaluated ENMO and MAD by computing their F1 score and precision as performance metrics. This assessment included wrist signal data from all 23 subjects with predictions available from the lower-back model.

3.9 Results

The results revealed that both ENMO and MAD classifications performed poorly in distinguishing walking from non-walking periods. Among the thresholds tested, 0.07g yielded the best relative performance for ENMO, whereas 0.05g provided the best results with MAD. We first obtained subject-level results, where the F1 and precision scores varied significantly across subjects, with values ranging from 0 to 0.35. We then also compute aggregate performance metrics across all subjects. For ENMO classification, the aggregate F1 Score was 0.03, and the aggregate Precision: 0.02. For MAD classification, the aggregate F1 Score was 0.04, and the aggregate Precision was 0.02. These findings suggest that while ENMO and MAD are straightforward heuristic methods, they are not sufficient for accurate activity classification using wrist-based accelerometer data, particularly when validated against lower back sensor ground truth labels.

Final Wrist Model - DCNN

3.10 Analytic Methods

To improve upon our baseline activity classification models, we implemented a self-supervised learning (SSL) approach by using the lower back gait detection algorithm proposed by Ionescu et. al to generate pseudo-labels for our wrist sensor data. This allowed our model to learn features specific to wrist data while transferring knowledge from the lower back domain, making the approach efficient in both time and cost.

To implement this, we fine-tuned a pretrained DCNN, “HarNet”, by Anandha Praba and Suganthi (2023) to our wrist data (OxWearables, n.d.). As shown in the architecture in Figure 11, convolutional neural networks (CNN) are used to capture spatial dynamics and long short-term memory (LSTM) layers handle the temporal, sequential aspects of our time series data. Batch normalization allows for more balanced layer inputs to speed up training while maintaining stability. ReLU activations introduce the much needed non-linearity into the model and prevent vanishing gradients. An Adam optimizer is used to efficiently update the network weights by adaptively adjusting the learning rate for each parameter. Finally, the fully connected layers act as classifiers, mapping the extracted features to our target activity classes.

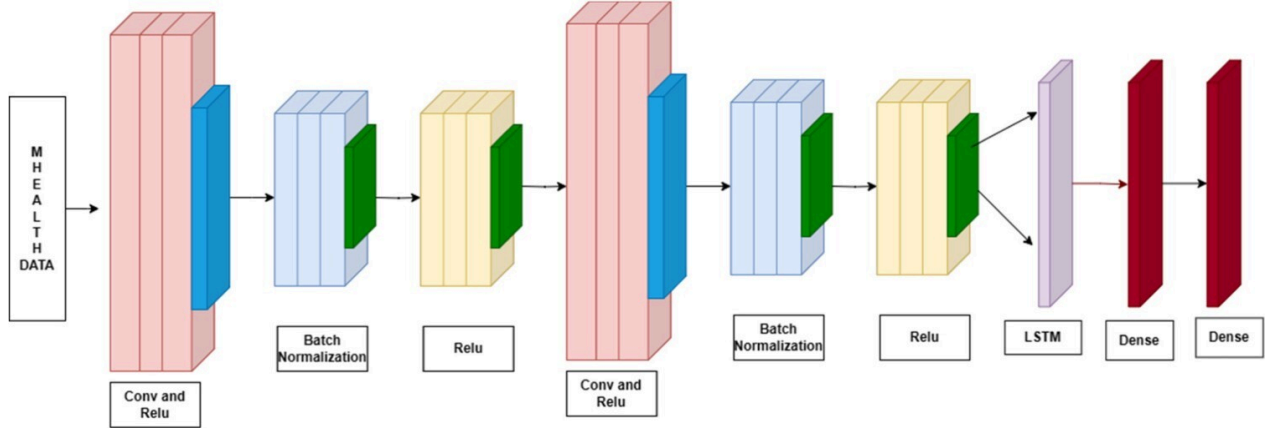


Fig. 11: DCNN process

Our fine-tuning strategy addressed significant class imbalance in the data, where 96% of the wrist signal data was mapped to the sedentary activity label (class 0). To mitigate this, we applied oversampling techniques, including SMOTE and random oversampling with added noise. Given our goal to prioritize precision, focusing on the model's ability to accurately detect the positive class (e.g., walking instances), we implemented a custom loss function that penalizes false positives more heavily. Additionally, we experimented with the following DCNN fine-tuning approaches: freezing the first residual block, freezing all convolutional layers, and incorporating adapter layers. For comparison, we also evaluated a version of the model without any fine-tuning modifications.

3.11 Validation and Testing

To evaluate and optimize the performance of our fine-tuned HarNet30 model, we split the dataset into training, validation, and test sets at the subject level to ensure no data leakage. Specifically, we allocated 60% of subjects for training, 20% for validation, and 20% for testing. This subject-level split ensured that the model was validated and tested on unseen participants, thereby assessing its generalizability.

During training, the model's validation precision was monitored at each epoch to track improvements. A custom training function implemented early stopping, halting the training process if no further improvements in validation precision were observed over a set number of epochs. This approach helped prevent overfitting while ensuring that the model prioritized precision.

The final model's performance on the test set was evaluated using standard classification metrics, including accuracy, F1-score, and confusion matrices, with a particular emphasis on precision for the positive class.

3.12 Results

Our evaluation demonstrated that applying random oversampling with noise significantly improved the model's performance across all metrics on the test set. Likewise, the custom precision loss function allowed us to identify walking bouts with high confidence. Consequently, both were used when comparing the different fine-tuning approaches of the DCNN.

Among the fine-tuning strategies, the no fine-tuning approach slightly outperformed the others, achieving an F1 score of 0.93 and a precision of 0.64. The next best-performing model was the one that froze the first residual block, which yielded an F1 score of 0.92 and precision of 0.64. The table below summarizes the evaluation metrics for each fine-tuning approach:

Position	Fine Tuning Approach	F1	Precision	# of Subjects
Lower Back	No Fine Tuning	0.93	0.64	23
	Freezing the First Residual Block	0.92	0.64	23
	Adapter Layers	0.93	0.61	23
	Freezing Convolutional Layers	0.81	0.15	23

Fig. 12: Evaluation Metrics for SSL Wrist Model Variations

These results suggest that we can confidently glean insights into an individual's activity using wrist accelerometer data. Moreover, the DCNN with transfer learning substantially outperformed the ENMO and MAD baseline methods. The superior performance of the DCNN highlights its ability to capture complex spatial and temporal patterns in wrist data that simple heuristic methods like ENMO and MAD fail to model effectively.

We could also further design new self-supervised tasks like predicting temporal shifts, scaling, or masked signals to improve the model's understanding of wrist accelerometer data. Likewise, different architectures could be explored as well as different optimizers and adaptive learning rate strategies.

Building upon the activity classification results, the next step in our analysis focused on predicting gait parameters, particularly walking speed, leveraging both lower-back and wrist-based accelerometer data.

Gait Speed

3.13 Analytic Methods - Lumbar

In order to calculate speed, the algorithm we used first requires stride length and cadence in the Mobilise-D pipeline.

Stride Length

The pipeline we built relies on a paper from Zijlstra and Hof (2003) that similarly attempts to predict gait characteristics from 3D-acceleration signals from lower-back sensors. The stride

length calculations in this paper are based on the mechanics of the body during walking, specifically the movement of the center of mass (CoM) in the sagittal plane (front-to-back). The authors used an inverted pendulum model to estimate stride length from changes in the vertical position of the CoM.

When walking, the CoM follows a circular trajectory as the body transitions through single-support phases. During each step, the height of the CoM rises and falls, creating a predictable arc-like movement. The authors assumed that the body behaves like a compass gait, where the legs act as pendulums, and step length depends on how much the CoM rises and falls during a stride.

Building on this, Soltani et al. (2021) introduced enhancements to the algorithm to address practical challenges in real-world data collection, such as missing or noisy initial contact (IC) detections. The Mobilise-D implementation integrates additional preprocessing and postprocessing steps, including outlier removal via Hampel filtering and linear interpolation for missing data. Furthermore, stride length is calculated as the doubled step length and is reported as an average value within one-second bins, providing temporal resolution suitable for gait analysis.

Following this foundation, the Mobilise-D algorithm (Mobgap Documentation, n.d.) works by first using a Madgwick filter on the raw accelerometer signals to estimate the orientation of the lower-back sensor over time (Madgwick, 2010). The signal is then high-pass filtered using a Butterworth filter (Selensnick & Burrus, 1998) with a 0.1 Hz cutoff to isolate vertical acceleration. This vertical acceleration is then integrated to compute vertical speed which is once again high-pass filtered at 1 Hz to mitigate drift. The velocity is integrated to derive vertical displacement ($d(t)$) to help represent the vertical motion of the CoM. Step length is estimated using the aforementioned inverted pendulum model with the equation:

$$l_{step} = A \cdot \sqrt{LBh \cdot d_{step}}$$

In this equation, A is a tuning coefficient that was found using grid search and LBh is the sensor height relative to the CoM. A Hampel filter (Pearson et al., 2015) is used to smooth the step lengths and remove outliers, while linear interpolation accounts for missing values. Finally, the stride length is calculated by doubling the smoothed step length and averaging it over one-second intervals.

Cadence

Following the Mobilise-D methodology (Mobgap Documentation, n.d.), cadence is estimated by calculating the number of steps taken per second using the initial contacts detected during the gait sequence. Initially, the step time is derived from the initial contacts, resulting in one fewer step time than the number of initial contacts. To address this, the last step time is replicated to match the total number of initial contacts, ensuring a cadence value is available for the final

contact in the gait sequence. Next, the step times are smoothed using a Hampel filter to remove any outliers. The average step time per second is then calculated, with special attention given to seconds that may not contain any initial contacts. For seconds with initial contacts, the average step time is computed, while for seconds without initial contacts, linear interpolation is used to estimate the step time based on surrounding values. If the gap between initial contacts exceeds a specified maximum of 3 seconds, the missing data is filled with NaNs. Gaps at the start and end of the recording are not interpolated, assuming that the gait sequence begins and ends with valid initial contacts. The step time per second is further smoothed to remove any residual outliers. Finally, the cadence is estimated by taking the inverse of the step time per second, providing a measure of steps taken per second.

Speed

Putting these two steps together, the pipeline (Mobgap Documentation, n.d.) performs a simple calculation to estimate the walking speed:

$$walking\ speed = \frac{stride\ length \times cadence}{60 \times 2}$$

Stride length is measured in meters, and cadence is initially in steps per minute. To convert cadence into steps per second, we divide by 60, and then divide by 2 to obtain strides per second. Multiplying the cadence in strides per second by the stride length in meters gives the walking speed in meters per second.

3.14 Analytic Methods - Wrist

Given ordinary least squares (OLS) tends to work well for short term prediction in high frequency time series data, this was our first choice method for wrist gait speed prediction despite the blatant assumption violations. In a self supervised approach, all wrist labels and two explanatory variables were derived from the lower back model algorithm's outputs, specifically from Mobilise-D's Ionescu model (Paraschiv-Ionescu et al., 2019). The wrist gait speed in meters per second is modeled as a linear function of derived cadence steps per minute and stride length, as well as the wrist sensor's raw acceleration X, Y, Z values.

3.15 Validation and Testing

The wrist speed model was trained only on the first subject's data, excluding the last 50,000 observations to ensure an independent test set for validation within that subject as well. This resulted in using 3,864,336 observations (each a 10 second interval) to fit the model. The model was used to predict gait speed for the test set of the first subject and then the remaining 22 subjects.

3.16 Results

OLS Regression Results

Dep. Variable:	walking_speed_mps	R-squared:	0.969
Model:	OLS	Adj. R-squared:	0.969
Method:	Least Squares	F-statistic:	2.380e+07
Date:	Sun, 01 Dec 2024	Prob (F-statistic):	0.00
Time:	19:16:59	Log-Likelihood:	8.5175e+06
No. Observations:	3864336	AIC:	-1.704e+07
Df Residuals:	3864330	BIC:	-1.703e+07
Df Model:	5		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-0.5159	0.000	-5038.637	0.000	-0.516	-0.516
accel_x	-0.0006	2.77e-05	-22.226	0.000	-0.001	-0.001
accel_y	0.0026	2.91e-05	90.665	0.000	0.003	0.003
accel_z	0.0022	2.84e-05	78.462	0.000	0.002	0.002
cadence_spm	0.0062	8.3e-07	7433.513	0.000	0.006	0.006
stride_length_m	0.6891	7.66e-05	8995.479	0.000	0.689	0.689

Omnibus:	1392122.311	Durbin-Watson:	0.012
Prob(Omnibus):	0.000	Jarque-Bera (JB):	66239974.654
Skew:	-0.992	Prob(JB):	0.00
Kurtosis:	23.186	Cond. No.	799.

Fig. 13: OLS Regression Model Output for Wrist Predicted Gait Speed

Figure 13 shows the fit of the OLS regression. The adjusted R^2 is again very high at 0.969, as is the F-statistic with a p-value of approximately zero, showing strong significance of the model overall. However, both the Omnibus and Jarque-Bera (JB) tests with p-values close to zero, indicating strong deviation from normality of the residuals. The data is skewed left, highlighted by a Skew of -0.992 and a Kurtosis of 23.186 which suggests a tail heavy distribution. Autocorrelation is also clearly present in the residuals given the Durbin-Watson statistic of 0.612. This is not surprising. Given it is time series data, the model is not set up to reflect the time series behaviors and patterns. Seeing Condition Number of 799 also points to multicollinearity in the explanatory variables, leading to unstable coefficients and leaving them unreliable for inference. Though there are clear violations of linear regression assumptions in terms of normality of residuals, linearity, independence of residuals, and uncorrelated independent variables, the model has strong prediction performance.

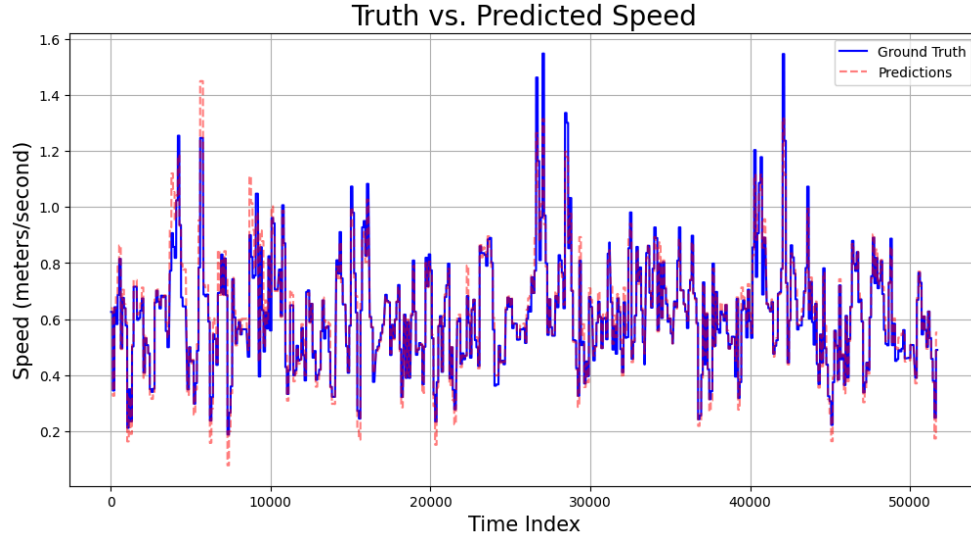


Fig. 14: Predicted Wrist Gait Speed vs. Ground Truth

The OLS regression generalized well on testing data, with an average adjusted R^2 of 0.960 across participants. This means that the majority of the variability in wrist walking speed can be predicted by this model. Likewise, the low average mean absolute error (MAE) of 0.02 and average root mean squared error (RMSE) of 0.03 highlights the model's prediction precision.

4 Discussion

This study investigated the feasibility of using wrist-worn IMUs to characterize spatiotemporal gait parameters, comparing their performance against a ground truth derived from lower back sensors. Our findings underscore the potential of wrist-based sensors in real-world gait analysis scenarios and illuminate key areas that warrant further optimization.

A primary achievement of this work was demonstrating that wrist-worn IMUs could capture essential gait dynamics with sufficient accuracy, as evidenced by strong performance in GSD and multiple other modalities. This outcome is significant because it points to a shift in gait analysis: the reliance on convenient and non-intrusive wrist-worn devices, as opposed to traditional, more cumbersome sensor placements like the lower back sensor. Such a shift not only broadens the contexts in which gait monitoring can occur—ranging from clinical assessments to at-home settings—but also has the potential to support continuous, convenient tracking that can inform proactive healthcare decisions.

A critical technical contributor to these results was adopting a custom precision-weighted loss function, combined with oversampling the minority class. This deliberate approach to algorithmic refinement enhanced the DCNN's robustness and sensitivity, particularly for noisy gait patterns. By balancing class distributions and accentuating subtle gait signals caused by high activity, our methodology allowed the model to learn more nuanced patterns and, consequently, yielded better generalization to diverse high activity gait patterns. This work, therefore,

highlights the importance of targeted methodological enhancements in addressing common challenges such as class imbalance, ultimately improving the reliability of machine learning models in wearable sensor applications.

Despite these positive outcomes, the study faced certain limitations that must be addressed in future research. The most pressing issue was the handling of missingness in the ground truth data. Although the lower back sensor provided a strong reference standard, real-world data collection inevitably entails periods of signal dropout, leading to substantial missingness. While imputation methods exist to fill these gaps, such strategies inherently risk introducing systematic bias. Unchecked, this bias can systematically distort performance metrics, ultimately undermining the credibility of subsequent model evaluations. A rigorous approach to missingness with probabilistic imputation techniques will be necessary to ensure the integrity of future analyses.

A second notable weakness emerged in the form of autocorrelation in the linear regression model used for wrist speed predictions. Although the current regression-based approach proved useful for prediction, the presence of autocorrelation limits its suitability for meaningful statistical inference and modeling interpretability. In practical terms, this means that while the model can forecast gait speed with reasonable accuracy, drawing deeper conclusions about underlying gait mechanics or sensor contributions remains elusive. Future investigations must consider more advanced modeling frameworks to better capture temporal dependencies and produce parameter estimates more aligned with the underlying gait biomechanics.

4.1 Future Work

Although the current DCNN approach performed effectively, exploring advanced architectures such as attention-based models or transformer-based approaches may offer even richer representations of gait dynamics. Fine-tuning hyperparameters via validation, improving training with different loss techniques, and integrating domain adaptation techniques could all further enhance generalizability and resilience to sensor placement variations.

The reliability of future work hinges on improving the robustness of the ground truth data. Integrating publicly available data generated from gait biomechanical studies could bolster the performance of the lower back model by further fine-tuning on this data. Such strategies could also mitigate the risks of low-quality ground truth recordings, leading to more meaningful model evaluations.

Future research could experiment with time-series-specific models to address autocorrelation and related modeling limitations in gait speed prediction. Techniques like recurrent neural networks, temporal convolutional networks, or ensembles incorporating both classical and modern machine learning methods may yield models that are simultaneously accurate and interpretable against autocorrelation artifacts. Furthermore, integrating different transfer learning approaches could

help models trained on one population or environment adapt more smoothly to new settings, making the technology more broadly applicable.

4.2 Conclusion

This study examined the feasibility of wrist-worn IMUs for gait analysis, challenging long-standing conventions tied to specialized sensor placements and complex equipment. By opening the door to more continuous monitoring of gait parameters, our work lays the groundwork for translating these tools into meaningful real-world interventions.

This study demonstrates that wrist-worn IMUs can be a practical platform for real-world gait analysis. They offer a more accessible alternative to traditional lower back sensor setups. By achieving robust performance in spatiotemporal gait parameter estimation, our work suggests that these more convenient devices can challenge existing paradigms and deliver continuous monitoring without requiring specialized equipment.

While our results are promising, key limitations must be addressed to ensure broader applicability. Missing ground truth data and autocorrelation issues in model inference highlight the importance of deliberate methodological enhancements. From applying precision-weighted loss functions to refining data preprocessing strategies and network architectures, targeted improvements are pivotal to ensuring both accuracy and reliability.

Our findings underscore that continued progress in modeling techniques, data handling, and validation methods will be essential for integrating wrist-based IMUs into clinical practice and everyday health assessment. As sensors and machine learning models become more sophisticated, these wearable systems will enable more proactive interventions, guiding clinicians toward evidence-based decisions on fall risk and rehabilitation strategies.

Ultimately, the trajectory of this field hinges on continuous innovation and interdisciplinary collaboration of the complexities of real-world data. By embracing these principles, we move closer to a future where wrist-worn IMUs become indispensable and reliable tools for enhancing patient outcomes and supporting population-level health initiatives.

4.3 Ethical Considerations

In this project, patient data was fully anonymized, ensuring that no personally identifiable information (PII) was collected, stored, or analyzed. Maintaining this level of privacy was paramount not only for compliance with ethical standards but also for sustaining the trust of individuals and communities who stand to benefit from these technologies. Beyond data handling, wearable gait analysis should not lead to discriminatory practices or exacerbate health disparities. By adhering to transparent data governance, informed consent, and equitable access, we aim to support clinical and community-oriented decision-making that respects personal autonomy, protects confidentiality, and promotes fair outcomes.

Sources

- Adamowicz, L., Christakis, Y., Czech, M. D., & Adamusiak, T. (2022). Scikit digital health: Python package for streamlined wearable inertial sensor data processing. *JMIR mHealth and uHealth*, 10(4), e36762.
- Anandha Praba, R., & Suganthi, L. (2024). HARNet: automatic recognition of human activity from mobile health data using CNN and transfer learning of LSTM with SVM. *Automatika: časopis za automatiku, mjerenje, elektroniku, računarstvo i komunikacije*, 65(1), 167-178.
- Bakrania, K., Yates, T., Rowlands, A. V., Esliger, D. W., Bunnewell, S., Sanders, J., ... & Edwardson, C. L. (2016). Intensity thresholds on raw acceleration data: Euclidean norm minus one (ENMO) and mean amplitude deviation (MAD) approaches. *PloS one*, 11(10), e0164045.
- Brand, Y. E., Schwartz, D., Gazit, E., Buchman, A. S., Gilad-Bachrach, R., & Hausdorff, J. M. (2022). Gait detection from a wrist-worn sensor using machine learning methods: A daily living study in older adults and people with Parkinson's disease. *Sensors*, 22(18), 7094.
- Femiano, R., Werner, C., Wilhelm, M., & Eser, P. (2022). Validation of open-source step-counting algorithms for wrist-worn tri-axial accelerometers in cardiovascular patients. *Gait & posture*, 92, 206-211.
- Iluz, T., Gazit, E., Herman, T., Sprecher, E., Brozgol, M., Giladi, N., ... & Hausdorff, J. M. (2014). Automated detection of missteps during community ambulation in patients with Parkinson's disease: a new approach for quantifying fall risk in the community setting. *Journal of neuroengineering and rehabilitation*, 11, 1-9.
- Kluge, F., Brand, Y. E., Micó-Amigo, M. E., Bertuletti, S., D'Ascanio, I., Gazit, E., ... & Mueller, A. (2024). Real-World Gait Detection Using a Wrist-Worn Inertial Sensor: Validation Study. *JMIR formative research*, 8, e50035.
- Madgwick, S. (2010). An efficient orientation filter for inertial and inertial/magnetic sensor arrays. *Report x-io and University of Bristol (UK)*, 25, 113-118.
- Mobgap Documentation. (n.d.). *mobgap.cadence.CadFromIc* (v0.5.1). Retrieved December 16, 2024, from <https://mobgap.readthedocs.io/en/v0.5.1/modules/generated/cadence/mobgap.cadence.CadFromIc.html>

Mobgap Documentation. (n.d.). *mobgap.stride_length.SlZijlstra* (v0.5.1). Retrieved December 16, 2024, from https://mobgap.readthedocs.io/en/v0.5.1/modules/generated/stride_length/mobgap.stride_length.SlZijlstra.html

Mobgap Documentation. (n.d.). *mobgap.walking_speed.WsNaive* (v0.5.1). Retrieved December 16, 2024, from https://mobgap.readthedocs.io/en/v0.5.1/modules/generated/walking_speed/mobgap.walking_speed.WsNaive.html#mobgap.walking_speed.WsNaive

OxWearables. (n.d.). *ssl_tutorial.ipynb* [Python notebook]. GitHub. Retrieved December 16, 2024, from https://github.com/OxWearables/Oxford_Wearables_Activity_Recognition/blob/master/6_self_supervised_learning/ssl_tutorial.ipynb

Paraschiv-Ionescu, A., Newman, C. J., Carcreff, L., Gerber, C. N., Armand, S., & Aminian, K. (2019). Locomotion and cadence detection using a single trunk-fixed accelerometer: validity for children with cerebral palsy in daily life-like conditions. *Journal of neuroengineering and rehabilitation*, 16, 1-11.

Pearson, R. K., Neuvo, Y., Astola, J., & Gabbouj, M. (2015, August). The class of generalized hampel filters. In *2015 23rd European signal processing conference (EUSIPCO)* (pp. 2501-2505). IEEE.

Rochester, L., Mazzà, C., Mueller, A., Caulfield, B., McCarthy, M., Becker, C., ... & Mobilise-D Consortium. (2020). A roadmap to inform development, validation and approval of digital mobility outcomes: the Mobilise-D approach. *Digital biomarkers*, 4(Suppl. 1), 13-27.

Selesnick, I. W., & Burrus, C. S. (1998). Generalized digital Butterworth filter design. *IEEE Transactions on signal processing*, 46(6), 1688-1694.

Small, S. R., Chan, S., Walmsley, R., von Fritsch, L., Acquah, A., Mertes, G., ... & Doherty, A. (2023). Development and validation of a machine learning wrist-worn step detection algorithm with deployment in the UK biobank. *MedRxiv*.

Soltani, A., Aminian, K., Mazza, C., Cereatti, A., Palmerini, L., Bonci, T., & Paraschiv-Ionescu, A. (2021). Algorithms for walking speed estimation using a lower-back-worn inertial sensor: A cross-validation on speed ranges. *IEEE transactions on neural systems and rehabilitation engineering*, 29, 1955-1964.

Soltani, A., Dejnabadi, H., Savary, M., & Aminian, K. (2019). Real-world gait speed estimation using wrist sensor: A personalized approach. *IEEE journal of biomedical and health informatics*, 24(3), 658-668.

Yuan, H., Chan, S., Creagh, A. P., Tong, C., Acquah, A., Clifton, D. A., & Doherty, A. (2024). Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. *NPJ digital medicine*, 7(1), 91.

Zijlstra, W., & Hof, A. L. (2003). Assessment of spatio-temporal gait parameters from trunk accelerations during human walking. *Gait & posture*, 18(2), 1-10.